

Consumers Underuse Co-Branded Credit Cards Outside of Their Featured Brands

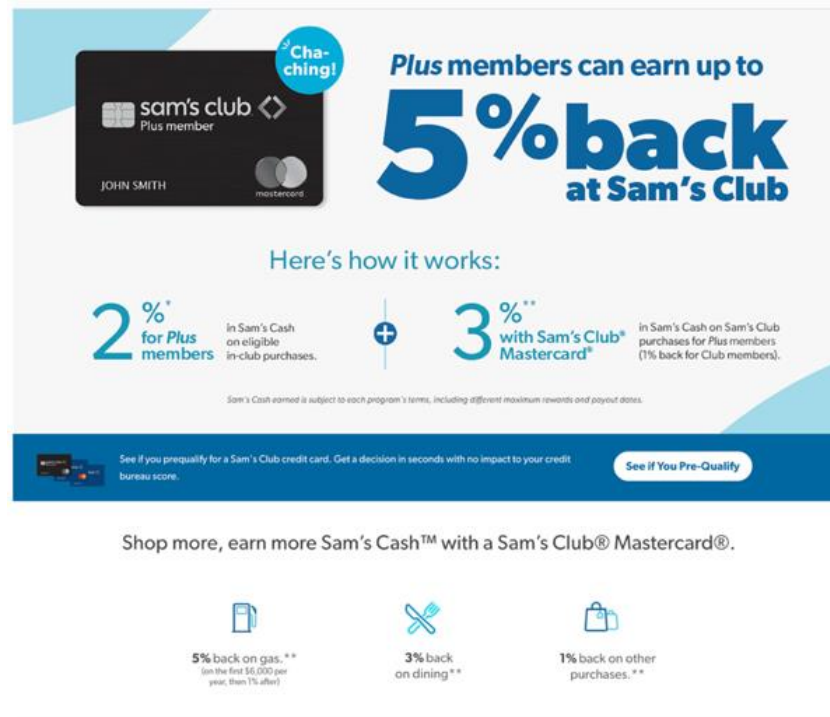
Web Appendix

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1. Sam's Club Mastercard Advertising Example

Figure S1. An Example of How Co-Branded Credit Cards Are Advertised



Source: <https://web.archive.org/web/20230608043936/https://www.samsclub.com/content/credit>

2. Study 1: Additional Information and Analyses

2.1. Study 1 - Pilot

To get an initial sense of how people use their co-branded and standard credit cards in real life, we surveyed 201 U.S. adults who possessed at least one co-branded credit card and one standard credit card.

Procedures

We aimed to recruit 200 U.S. adults who possessed at least one co-branded credit card and one standard bank credit card. To find qualified participants, we prescreened 701 participants on Prolific.com, asking them to report the number of co-branded and standard bank credit cards they held. Three hundred eight participants possessed at least one co-branded card and one standard bank credit card and thus were invited to participate in the pilot study, which we set to enroll 200 participants on a first-come first-serve basis. Two hundred one participants completed the pilot study ($M_{\text{age}} = 40.5$; 54% female, 45% male, and 1% non-binary or preferring not to say).

In the pilot study, participants first indicated the full names of one co-branded credit card and one standard bank credit card that they used most often by selecting from two drop-down lists, one list for standard bank credit cards and the other for co-branded credit cards. If their

cards were not in the lists, they were asked to type out the full names of their cards in text-entry boxes. For each card, participants responded to seven questions. We share the full survey and all responses in our online repository: <https://researchbox.org/4109>.

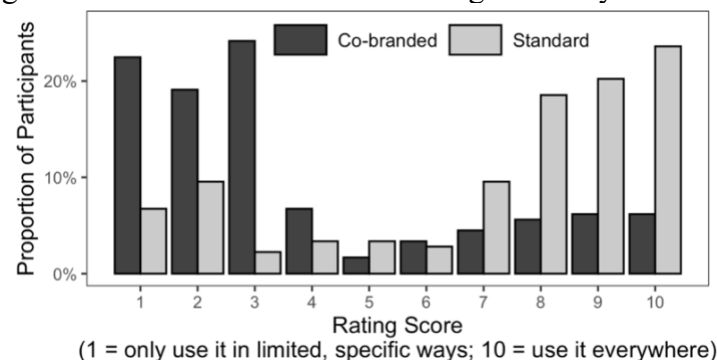
Results

Because participants indicated the precise names of their cards, we were able to verify that almost all of their co-branded credit cards (99%) offered purchase-based rewards outside of the featured brands that were roughly comparable to their standard bank credit cards, as summarized in Web Appendix 2.2.

For each card, participants answered the following question, which serves as our dependent variable: “Do you tend to use this card for specific types of purchases or at specific stores, or do you tend to use this card everywhere? Please rate your card usage on a scale from 1 (only use it in limited, specific ways) to 10 (use it everywhere).”

Participants reported markedly distinct patterns of use between the two types of credit cards. As shown in Figure S2, participants generally used their standard credit cards to make a variety of purchases but tended to use their co-branded credit cards in limited, specific ways. On average, participants reported substantially more restrictive usage for their co-branded credit cards than their standard bank credit cards ($M_{\text{co-branded}} = 3.83$, $SD = 2.88$; $M_{\text{standard}} = 7.13$, $SD = 2.96$; $t(177) = 9.79$, $p < .001$).

Figure S2. Breadth of Credit Card Usage in Study 1 Pilot



This pattern offers suggestive evidence that consumers use the two types of credit cards differently, consistent with our hypothesis that people underuse their co-branded credit cards outside of the cards’ featured brands. Next, in the main study, we further examine this pattern by examining real credit card transactions documented on credit card statements.

2.2. Study 1 - Pilot: Credit card rewards

Because participants indicated the precise names of their cards, we were able to identify all reported credit cards and verify their reward structures. Almost all of their cards offered purchase-based rewards: 99% of the co-branded credit cards and 88% of the standard bank credit cards offered at least some base-tier rewards everywhere the cards were accepted, which were typically 1% of the purchase amounts or 1 reward point per dollar spent. Beyond the base-tier rewards, similar to their standard bank credit cards, most of their co-branded credit cards offered higher-tier rewards on certain spending categories even outside their featured brands, such as travel or dining. These category-specific elevated rewards were higher in amounts than the respective card’s universal rewards, such as 2% or more of the purchase amounts or 2 points or more per dollar spent. Specifically, 89% of the co-branded credit cards offered elevated rewards

on certain spending categories outside their featured brands, while 70% of the standard bank credit cards offered such higher rewards. Table S1 summarizes the elevated rewards of both types of cards. We reported the reward terms of each credit card of each participant in the data file available in our online repository: <https://researchbox.org/4109>.

Table S1. Proportions of Credit Cards Offering Elevated Rewards for Different Categories of Purchases in Study 1's Pilot

	Standard bank credit cards	Co-branded credit cards
Groceries	22%	11%
Gasoline	15%	67%
Restaurants	29%	76%
Travel	41%	54%
Local transit	1%	37%
Telecom	8%	6%
Others	34%	18%

Notes: The reward information was collected from each credit card's official website in February 2024.

2.3. Study 1 - Pilot: Reasons for having the credit cards

Table S2. Reasons for Having The Credit Cards

Reasons	% Indicating that reason	
	Co-branded credit card	Standard credit card
Being pre-qualified	0.5%	0%
Better credit limit	0.5%	0%
Better services	0.5%	0.5%
Brand liking	15.9%	0%
Co-branded/Specific rewards	48.3%	9.0%
Convenience of payment	10.9%	14.4%
General rewards	28.4%	36.3%
Low APR	2.5%	6.5%
Low introductory APR	0.5%	0%
Pay-over-time features	11.4%	14.4%
Sign-up bonus	29.4%	27.4%
To build credit scores	1.5%	0.5%
To track certain types of spending	1.0%	2.0%
To try to get their very first credit card	0.5%	0%
Other	0%	4%

2.4. Study 1 - Main Study: Compliance with credit card submission requirements

As illustrated in Table S3, most participants (126 out of the 153) provided the transaction images of both a valid co-branded credit card and a valid standard bank credit card as required. However, because we were unable to monitor submissions in real time, some participants erroneously provided 0 or 2 valid cards for each type. The system had a basic mechanism that prohibited them from providing more than two cards in total. Invalid credit cards included debit cards and closed-loop store cards¹. In Study 1's analysis, we focus on the 126 fully compliant participants and their reported credit cards.

Table S3. Number of valid credit cards each participant provided

		Valid co-branded credit card(s)		
		0 card	1 card	2 cards
Valid standard bank credit card(s)	0 card	5 Participants	4 Participants	1 Participant
	1 card	13 Participants	126 Participants	N/A
	2 cards	4 Participants	N/A	N/A

Note. Each participant was asked to provide 1 co-branded credit card and 1 standard bank credit card. However, because we were unable to monitor their submissions in real time, a few non-compliant participants provided 0 or 2 cards for each type. Participants could not provide more than 2 cards in total.

2.5. Study 1 - Main Study: Credit card transaction coding details

As preregistered, we first attempted to code each transaction into one of the 15 categories based on the type of purchase. For example, the transaction “MCDONALD'S M# OF VA” would be coded into the dining category. If the type of purchase was ambiguous based on the description, we would then try to code it based on its vendor type. For instance, the transaction “AMAZON MKTPL # Amzn.com/bill WA” could cover anything available on Amazon. Therefore, we coded it into the “e-commerce” category. All the other transactions went into the “Others” category.

2.6. Study 1 - Main Study: Credit card rewards details

Similar to the pilot, we were able to identify all reported credit cards from the 126 compliant participants and verify their reward structures based on the credit card names. Almost all of their cards offered purchase-based rewards: 96% of the co-branded credit cards and 81% of the standard bank credit cards offered at least some base-tier rewards everywhere the cards were accepted, which were typically 1% of the purchase amounts or 1 reward point per dollar spent. Beyond the base-tier rewards, similar to participants' standard bank credit cards, most of their co-branded credit cards offered higher-tier rewards (i.e., rewards higher than the card's rewards on general spending, such as 2% or more of the purchase amounts or 2 points or more per dollar spent) on certain spending categories even outside their featured brands, such as travel or dining.

¹ We also preregistered to exclude PayPal credit cards and Apple Cards because their brands focus on certain payment methods (PayPal and Apple Pay) that apply to all types of purchases. Yet, no participants reported these cards.

Specifically, 93% of the co-branded credit cards offered elevated rewards on certain spending categories outside their featured brands, while 69% of the standard bank credit cards offered such higher rewards. Table S4 summarizes the elevated rewards of both types of cards. We reported the reward terms of each credit card of each participant in the data file available in our online repository: <https://researchbox.org/4109>.

Table S4. Proportions of Credit Cards Offering Elevated Rewards for Different Categories of Purchases in the main study of Study 1

	Standard bank credit cards	Co-branded credit cards
Groceries	19%	15%
Gasoline	17%	63%
Restaurants	29%	79%
Travel	45%	62%
Local transit	0%	40%
Telecom	2%	6%
Others	28%	13%

Notes: The reward information was collected from each credit card's official website in March 2025.

2.7. Study 1 - Main Study: The NerdWallet conversion rates

Credit card rewards vary in form and value. Depending on the program, rewards may be paid as cash back, airline miles, hotel points, store credits, gift cards, statement credits, or other brand-specific perks. Because these rewards are not all denominated in directly comparable dollar terms, we converted them into dollar values for our analyses. Specifically, we first used the point valuations reported by NerdWallet in 2025 (Table S5). Then, for cards whose rewards were not covered by the NerdWallet point valuations but were redeemable in cash, such as the Costco Anywhere Visa and the Discover it Cash Back, we directly used the cash values of their points. These two valuation approaches together covered 87% of the credit cards in our dataset.

For the remaining cards, either their reward points had explicit dollar values toward in-store purchases, such as the REI Co-op Mastercard with each point worth 1 cent in store credits, or they did not offer any rewards, such as the Merrick Bank Credit Card. For the store-credit rewards, we used their face values in our analyses. We also conducted more conservative analyses using cards with cash rewards only and obtained the same results, as described in Section 2.10.

Table S5. NerdWallet Conversion Rates 2025

Reward Program	Point Value (cents)
Aeroplan (Air Canada)	1.1
Alaska Airlines Mileage Plan	1.3
American Airlines AAdvantage	1.6
American Express Membership Rewards	1.0
Avianca	1.6
Bank of America Travel Rewards	1.0
Barclaycard Arrival points	1
Best Western Rewards	0.6
Bilt Rewards	1.25
British Airways	1.2
Capital One miles	1.0
Chase Ultimate Rewards® (all other cards)	1.0
Chase Ultimate Rewards® (Preferred, Reserve, and Ink cards)	2.1
Choice Privileges	0.8
Citi ThankYou Points (all other cards)	1.0
Citi ThankYou Points (Premier card)	1.25
Delta Air Lines SkyMiles	1.2
Discover miles	1
Frontier Miles	1.5
Hawaiian Airlines HawaiianMiles	1.0
Hilton Honors	0.6
Hyatt (World of Hyatt)	0.5
JetBlue TrueBlue	1.5
KLM/Air France	0.8
Marriott Bonvoy	0.9
Southwest Airlines Rapid Rewards	1.3
Spirit Airlines Free Spirit	1.3
Turkish Airlines	0.7
U.S. Bank FlexPerks	1.5
United Airlines MileagePlus	1.2
Virgin Atlantic	1.4
Wells Fargo Go Far Rewards (all other cards)	1.0
Wells Fargo Visa Signature® Credit Card	1.5
Wyndham Rewards	1.2

Note: NerdWallet updates their valuations annually on their website: <https://www.nerdwallet.com/travel/learn/airline-miles-and-hotel-points-valuations>. The valuations shown in this table were captured in 2025.

2.8. Study 1 – Main Study: Quantifying Forgone Rewards Value

Using the NerdWallet conversion rates (see section 2.7), we computed the total dollar value each participant missed out on for each of their credit cards across the three billing cycles. On average, participants missed out on more rewards for their co-branded credit card than for their standard bank credit card ($M_{\text{co-branded}} = \9.60 , $SD = 22.9$; $M_{\text{standard}} = \$3.46$, $SD = 22.9$, $t(114) = 2.32$, $p = 0.022$).

2.9. Study 1 - Main Study: Logit Analysis

To test the robustness of the usage-constraining effect of co-branded credit cards, we estimated a fixed-effects logit with relevant covariates. We fitted the model with a subset of transactions that met two criteria: (A) each transaction was outside of the featured brand of that individual's co-branded card, and (B) each transaction would render different reward value if paid by different cards. We further excluded 65 transactions that did not display recognizable

transaction amounts, leaving a final sample of 6,064 transactions. We coded the dependent variable as 1 if a specific transaction was paid with the higher-reward credit card, and 0 otherwise. We included six regressors: (1) a co-branded indicator that equaled 0.5 if the higher-reward credit card was co-branded and -0.5 otherwise; (2) transaction amount in US dollars; (3) the reward gap between the higher and lower rewards for a specific transaction, measured in cents per dollar spent and mean-centered; (4) the interaction between the co-branded indicator and the reward gap; (5) individual fixed effects that captured all unobservables invariant within each individual; (6) spending category fixed effects that captured all unobservables invariant within each type of spending (e.g., groceries, fuel, dining, etc.).

As summarized in Table S6, controlling for all covariates, users were still significantly less likely to use their higher-reward cards for purchases if the higher-reward cards were co-branded with brands that did not sell the items being purchased ($b = -1.38$, $SE = .11$, $p < .001$). Additionally, although users were more likely to use their higher-reward standard bank credit card when the reward gap was larger ($b = .43$, $SE = .08$, $p < .001$), such sensitivity was substantially reduced when the higher-reward card was co-branded (interaction $b = -.90$, $SE = .15$, $p < .001$), as visualized in Figure S3. In other words, unlike standard bank credit cards, greater reward advantages failed to promote use of co-branded credit cards outside of their featured brands.

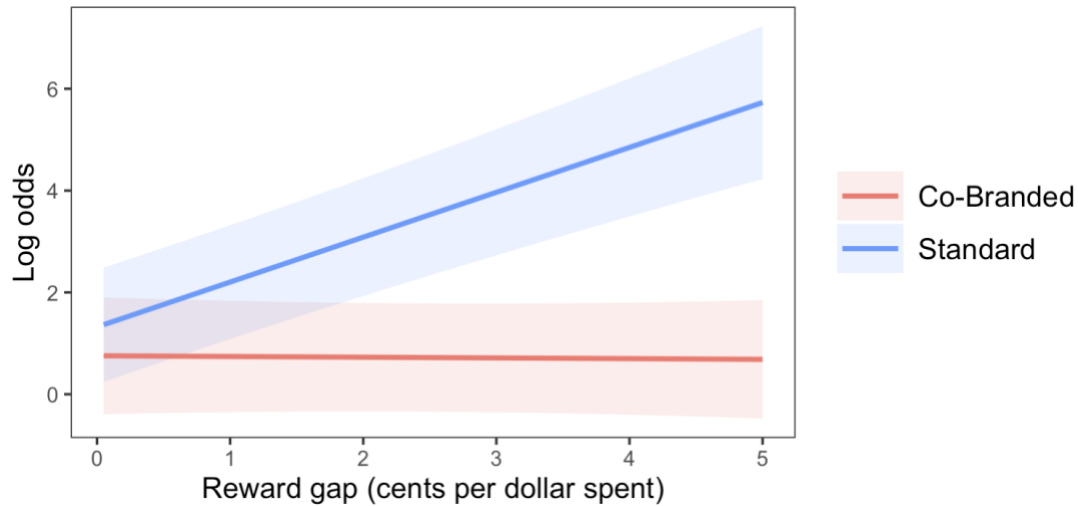
Table S6. Study 1 Logit Results

	<i>DV: 1 = Higher-reward card was used 0 = Lower-reward card was used</i>	
	Coefficient	<i>SE</i>
Co-branded indicator	-1.38***	(0.11)
Transaction amount	-0.02	(0.02)
Reward gap	0.43***	(0.08)
Co-branded indicator x Reward gap	-0.90***	(0.15)
Individual fixed effects	✓	
Spending category fixed effects	✓	

Notes:

1. $N = 6,064$.
2. pseudo- R^2 (McFadden's) = 0.45.
3. Transaction amount unit was rescaled to USD 100.
4. p -values: * < .05; ** < .01; *** < .001;

Figure S3. Interaction Plot: The Log Odds of Using the Higher-Reward Card



Notes: Ribbons represent 95% confidence bands. The figure illustrates the estimated interaction between credit card type (i.e., the co-branded indicator) and reward gap, holding the other variables constant at their reference levels.

As a more conservative test, we repeated the analyses using only cash-reward cards, whose rewards were directly comparable because they could be redeemed in cash. The same pattern of results persisted. We discuss these analyses in the next section.

2.10. Study 1 - Main Study: Repeating Study 1 Analyses with Cash-Reward Cards Only

As a more conservative test, we repeated Part 2 of Study 1's analyses with sixty-nine participants whose co-branded and standard credit cards both offered cash rewards and thus were directly comparable. The same pattern of results holds. As summarized in Table S7, when their co-branded credit card offered higher cash rewards, they used the co-branded credit cards only 33% of the time, reliably lower than when their standard credit cards offered higher rewards (81%, $\chi^2(1) = 765, p < .001$).

Table S7. Results with Participants whose Cards Both Offered Cash Rewards

Situation (which card offers the higher reward?)	Number of transactions	Percent of transactions where the higher-reward card was used
Standard credit card offers higher reward	2,084	81%
Co-branded card offers higher reward...		
• For purchases outside the brand	1,224	33%
• For purchases within the brand	1,117	98%
Both cards offer equal rewards	587	N/A
Total	5,012	

Then, we repeated the logistic regression as described in Appendix 2.9 with the subset of data from participants whose cards both offered cash rewards. We obtained the same pattern of results, as summarized in Table S8.

Table S8. Logit Results with Participants whose Cards Both Offered Cash Rewards

	DV: 1 = Higher-reward card was used 0 = Lower-reward card was used	
	Coefficient	SE
Co-branded indicator	-2.56***	(0.22)
Transaction amount	-0.06	(0.05)
Reward gap	0.29*	(0.12)
Co-branded indicator x Reward gap	-1.05***	(0.23)
Individual fixed effects	✓	
Spending category fixed effects	✓	

Notes:

1. N = 3,308.
2. pseudo-R² (McFadden's) = 0.46.
3. Transaction amount unit was rescaled to USD 100 for the ease of displaying its coefficient and SE.
4. *p*-values: * < .05; ** < .01; *** < .001;

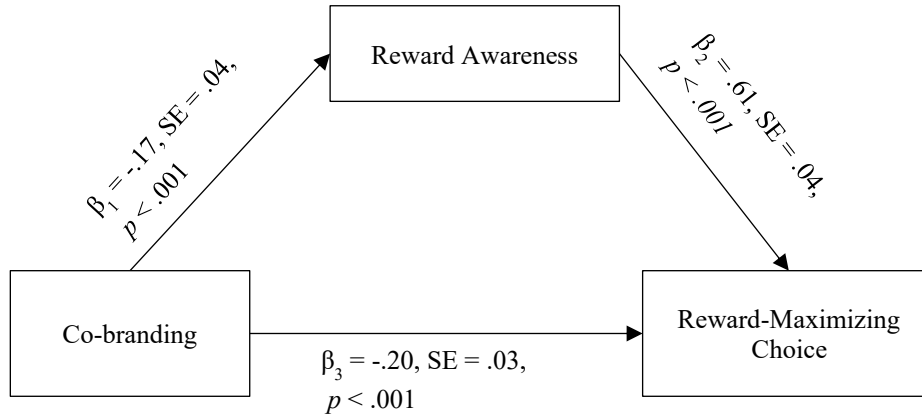
3. Study 2: Mediation through reward awareness

We fitted a mediation model to test whether the Best Buy co-branding discouraged participants from using the reward-maximizing card by reducing awareness of rewards outside of Best Buy. We coded the independent variable, co-branding, as 1 if the reward-maximizing card was the Best Buy Visa and 0 if it was the BOA Visa. Then, we coded the mediator, reward awareness, as 1 if participants correctly indicated which card earned more cash back on groceries, and 0 otherwise. Finally, we coded the dependent variable, reward-maximizing choice, as 1 if they chose the reward-maximizing card to pay for the groceries and 0 otherwise. As in the

main analyses, the mediation model was estimated using the “sem” command in the “lavaan” package in R. By default, the model was fitted using maximum likelihood estimation, and standard errors were estimated using the first-order Taylor approximation (i.e., the Delta method).

Our results showed a significant total effect of co-branding on the reward-maximizing choice ($\beta_{\text{total}} = -.31$, $SE = .04$, $p < .001$). This total effect was reliably mediated through reward awareness ($\beta_{\text{indirect}} = -.10$, $SE = .03$, $p < .001$). There remained a reliable direct effect ($\beta_{\text{direct}} = -.20$, $SE = .03$, $p < .001$), suggesting a separate process pathway, which we have shown to be the one via perceived card-purchase fit. Figure S4 illustrates the mediation.

Figure S4. Study 2 Mediation



4. Study 4: Individual “Fit” Measures

Participants indicated that the reward-maximizing card had a worse fit with the grocery purchase when it was co-branded with Best Buy ($M_{\text{fit, co-branded}} = 6.47$, $SD = 2.20$) than when it was a standard credit card ($M_{\text{fit, standard}} = 7.83$, $SD = 3.36$; $t(585) = 5.79$, $p < .001$). Similarly, they also indicated that the reward-maximizing card was less compatible with the grocery purchase when it was co-branded ($M_{\text{compatible, co-branded}} = 6.81$, $SD = 2.18$; $M_{\text{compatible, standard}} = 7.95$, $SD = 3.25$; $t(585) = 4.94$, $p < .001$).

5. Study 6A: Individual “Fit” Measures

Participants indicated that the Shell Mastercard fit with Shell gasoline the highest ($M_{\text{Shell}} = 4.43$, $SD = 1.07$), followed by car engine coolant from AutoZone ($M_{\text{coolant}} = 1.90$, $SD = 2.52$), movie tickets from AMC ($M_{\text{movie}} = .71$, $SD = 3.04$), and BP gasoline ($M_{\text{BP}} = .51$, $SD = 3.21$) in descending order. BP gasoline was considered a poorer fit with the Shell Mastercard than Shell gasoline ($t(120) = -9.05$, $p < .001$).

Similarly, participants indicated that the Shell Mastercard was most compatible with Shell gasoline ($M_{\text{Shell}} = 4.64$, $SD = 1.00$), followed by car engine coolant from AutoZone

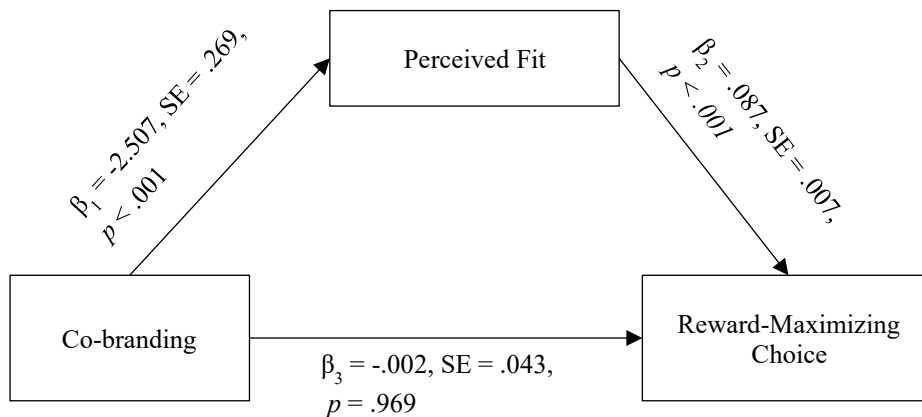
($M_{\text{coolant}} = 2.39$, $SD = 2.64$), movie tickets from AMC ($M_{\text{movie}} = 1.64$, $SD = 3.01$), and BP gasoline ($M_{\text{BP}} = .95$, $SD = 3.38$) in descending order. BP gasoline was considered less compatible with the Shell Mastercard than Shell gasoline ($t(120) = -8.17$, $p < .001$).

6. Study 6B: Mediation

We fitted a mediation model to test whether the Shell co-branding discouraged participants from using the reward-maximizing card by reducing the perceived card-purchase fit. We coded the independent variable, co-branding, as 1 if the reward-maximizing card was the Shell Visa and 0 if it was the Citi Visa. Then, we coded the dependent variable, reward-maximizing choice, as 1 if they chose the reward-maximizing card to pay for the gasoline and 0 otherwise. As in the main analyses, the mediation model was estimated using the “sem” command in the “lavaan” package in R. By default, the model was fitted using maximum likelihood estimation, and standard errors were estimated using the first-order Taylor approximation (i.e., the Delta method).

Our results showed a significant total effect of co-branding on the reward-maximizing choice ($\beta_{\text{total}} = -.219$, $SE = .046$, $p < .001$). This total effect was reliably mediated through perceived fit ($\beta_{\text{indirect}} = -.221$, $SE = .030$, $p < .001$). There was no significant direct effect ($\beta_{\text{direct}} = -.002$, $SE = .043$, $p = .969$).

Figure S5. Study 6B Mediation



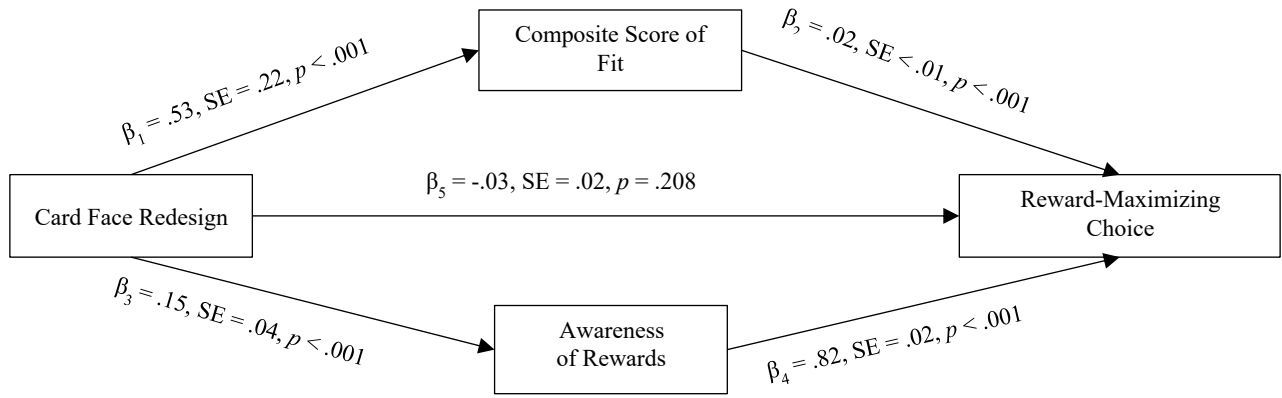
7. Study 7: Mediation

We tested, via preregistered parallel mediation, whether the card face design encouraged out-of-brand usage through increased awareness of its out-of-brand rewards and improved perceptions of card-purchase fit. We coded the independent variable, card face design, as 1 if the visual cues about broader card utility were present, and 0 otherwise. We coded one mediator, awareness, as 1 if participants correctly indicated which card earned more cash back on dining, and 0 otherwise. Then, we included the other parallel mediator, perceived fit, measured as participants’ ratings on the 0-10 scale. Finally, we coded the dependent variable, reward-maximizing choice, as 1 if they chose the Citi Shell Mastercard to pay for their dining and 0 otherwise. The estimation methods were the same as in Study 4.

Results showed that the effect of the redesign on card choice was reliably mediated through two separate pathways: increased awareness of the Shell card’s reward advantage on dining

($\beta_{\text{indirect, aware}} = .12, SE = .03, p < .001$) and the improved perceptions of card-purchase fit ($\beta_{\text{indirect, fit}} = .01, SE < .01, p = .042$). Figure S6 visualizes the mediation model and its estimates.

Figure S6. Card Face Redesign Increases Out-of-Brand Usage by Improving Reward Awareness and Perceived Fit



8. Supplemental Study A: Brand logos and the position of rewards

One concern with our experiments may be that a brand logo was featured only in the co-branded condition. For example, the terms table in Study 2 (see Figures 2a & 2b in the main text) included a Best Buy logo in the co-branded condition. Another concern may be that the position of the reward terms (at the very bottom of the table) may have contributed to a lack of awareness of reward terms. This supplemental study addresses both concerns by removing any brand logos and placing the reward terms at the top of the table.

Method

As preregistered (<https://aspredicted.org/cwks-w2rf.pdf>), we recruited 400 participants on Prolific.com. Participants were randomly assigned to either a Control condition or a Co-branded condition. In both conditions, participants imagined possessing two credit cards, and the terms of the two cards were displayed side by side (see Figures S7a & S7b). Notably, no brand logo was displayed for either card, and the reward terms were placed at the top of the table.

Figure S7a. Credit Card Information as Displayed to Participants – Control Condition

	Chase Visa	Citi Visa
Cashback Rewards	• 3% on gas and restaurant purchases. • 1% on all other purchases.	• 3% on online retail purchases. • 2% on all other purchases.
Credit Line	\$8,000	\$8,000
Current Balance (After most recent payment)	\$0	\$0
Interest Rate (APR)	19%	19%

Figure S7b. Credit Card Information as Displayed to Participants – Co-Branded Condition

	Chase Visa	Best Buy Visa by Citi
Cashback Rewards	• 3% on gas and restaurant purchases. • 1% on all other purchases.	• 3% on Best Buy purchases. • 2% on all other purchases.
Credit Line	\$8,000	\$8,000
Current Balance (After most recent payment)	\$0	\$0
Interest Rate (APR)	19%	19%

In the Control condition, participants imagined that they carried a Chase Visa and a Bank of America (BOA) Visa. These two credit cards had the same terms (the same credit line, current balance, and APR) except for their rewards. The Chase Visa paid 3% cash back at gas stations and restaurants and 1% on all other purchases. The BOA Visa paid 3% cash back on online retail purchases and 2% on all other purchases. In the Co-branded condition, participants imagined possessing the same Chase Visa and a Best Buy Visa. The Best Buy Visa was identical to the BOA Visa from the Control condition except for one item in its reward structure. The Best Buy Visa paid 3% cash back at Best Buy instead of on online purchases. All other attributes were identical to the BOA Visa from the Control condition including 2% cash back on all other purchases.

After learning about their available credit cards, participants proceeded to the next screen where they imagined that they were shopping for groceries. Once participants were ready to check out, they saw their total purchase amount and chose one of their two credit cards to use to make the payment. Then, as an attention check, all participants were asked to indicate the two credit cards that they had in the grocery store scenario by selecting one of three options: “Chase Visa and Best Buy Visa,” “Chase Visa and Bank of America Visa,” or “American Express and Citi Mastercard”. We preregistered a plan to exclude participants who did not pass this general attention check. Fourteen participants failed, leaving us a final sample of 386 participants ($M_{\text{age}} = 35.1$ years; 51% female, 46% male, and 3% indicating “Other”).

Results

As preregistered, we compared the proportion of participants choosing the reward-maximizing card between conditions. Consistent with Study 2 in the main text, participants were less likely to use the reward-maximizing credit card for groceries if it was co-branded with Best Buy: 62% in the Control condition, versus 45% in the Co-branded condition, chose to use the reward-maximizing card ($\chi^2(1) = 10.6, p = .001$). Even though no brand logo was displayed, and even though the reward information was placed at the top of the terms table, participants were still reluctant to use the reward-maximizing card if it featured a brand that was not associated with the purchase.

9. Supplemental Study B: Understanding the role of brand preference

A typical experimental design in this paper is to assign a co-branded credit card to participants. In the real world, however, people select their own co-branded credit cards. If some participants in the Co-branded condition disliked the co-brand, they might not want to use the card at all, whether the purchase was within or outside of the co-brand, potentially contributing to the pattern that we observed in the experiments. We address this concern in this supplemental study by conducting an experiment with actual co-branded credit card users and measuring their intention to use their own co-branded credit cards outside the featured brands.

Method

We preregistered to recruit 400 U.S. adults who possessed at least one of these credit cards: Walmart Rewards Mastercard, Target RedCard Mastercard, or Discover it Chrome (preregistration: <https://aspredicted.org/xjpw-g3fs.pdf>). We chose these three cards because they each offered 2% cash back on certain fuel purchases. (The details of the Walmart Rewards Mastercard cashback offer on fuel turned out to be more nuanced than we originally recognized, and we address this later.) We first prescreened 1705 participants based on whether they possessed any of the three cards. Four hundred eighty-seven possessed at least one of the three cards and thus were invited to participate in the main study, which was set to enroll 400 participants on a first-come first-serve basis. Four hundred participants completed the main study. At the beginning of the main study, participants indicated which of the three credit cards they had in real life. In contrast to what they indicated in the prescreening survey, six participants indicated that they did not have any of the three credit cards and were thus excluded. The remaining 394 participants proceeded.

As preregistered, participants who indicated they had at least one of the co-branded credit cards (Target or Walmart) but did not have the Discover card were assigned to the Co-branded condition. Those who indicated they had only the Discover card were assigned to the Control condition. Those who had both the co-branded card(s) and the Discover card were randomly assigned to one of the two conditions. As a result, 176 participants were assigned to the Control condition and 213 to the Co-branded condition.

Participants imagined that they carried two credit cards for everyday purchases. Those in the Control condition imagined they carried a Chase Freedom Unlimited Visa and a Discover it Chrome card. Those in the Co-branded condition imagined that they carried a Chase Freedom Unlimited Visa and a co-branded credit card that they actually possessed in real life—either the Walmart Rewards Mastercard or the Target RedCard Mastercard. If a participant in the Co-branded condition indicated that they had both the Walmart and the Target cards, that participant would be randomly assigned one of the two cards. Participants were then provided a table containing the credit card terms, as shown in Figures S8a through S8c. For each credit card, the table displayed its actual name and card appearance, followed by a hypothetical credit line, a hypothetical statement balance, a hypothetical APR, and its real reward structure. The credit line and statement balance were the same for all cards both within a condition and between conditions. Specifically, the statement balance was set to zero to minimize concerns about interest charges. The APR was set at the midpoint of the card's actual APR range. Both the APR and reward information were obtained from each credit card's official website when the study was designed in May 2024.

Participants were guided through a hypothetical process of getting gasoline at a gas station. Following the normal sequence for paying with credit at the gas station, participants were first asked to select a card to pay with, then select the type of fuel, and then decide how many gallons they wanted to pump. In both conditions, the Chase Freedom Unlimited Visa would offer a lower reward on gasoline (1.5% cash back) than the other card, whether it was the Walmart Rewards Mastercard (2% cash back), the Target RedCard Mastercard (2% back in Target rewards), or the Discover it Chrome (2% cash back). The Target rewards could only be used at Target, equivalent to Target store credits or a Target gift card. To understand whether the Target RedCard Mastercard holders considered the Target rewards as comparable to cash, we further asked them whether one dollar in Target gift card was worth less than, equal to, or more than a dollar in cash.

Lastly, as an attention check, participants answered a multiple-choice question by selecting the pair of credit cards they had in the hypothetical scenarios. Five participants did not answer correctly and were excluded, as preregistered, leaving us with a final sample of 389 participants ($M_{age} = 39.7$ years; 63% female, 35% male, and 2% non-binary or preferring not to answer).

Figure S8a. Credit Card Terms in the Control Condition

 Chase Freedom Unlimited		 Discover it	
Credit Line	\$8,000	Credit Line	\$8,000
Statement Balance	\$0	Statement Balance	\$0
Purchase APR	24.86%	Purchase APR	22.74%
Cash Back Rewards	• 3% on dining at restaurants, including takeout and eligible delivery services. • 3% on drugstore purchases. • 5% on travel purchased through Chase Travel. • 1.5% on all other purchases.	Cash Back Rewards	• 2% at gas stations. • 2% at restaurants. • 1% on all other purchases.

Figure S8b. Credit Card Terms in the Co-Branded Condition—Walmart

	 Chase Freedom Unlimited	 Walmart Rewards® Card (Mastercard)
Credit Line	\$8,000	\$8,000
Statement Balance	\$0	\$0
Purchase APR	24.86%	24.74%
Cash Back Rewards	• 3% on dining at restaurants, including takeout and eligible delivery services. • 3% on drugstore purchases. • 5% on travel purchased through Chase Travel. • 1.5% on all other purchases.	• 5% at Walmart.com, including pickup and delivery. • 2% in Walmart stores and fuel stations, at restaurants and on travel. • 1% Everywhere else.

Figure S8c. Credit Card Terms in the Co-Branded Condition—Target

	 Chase Freedom Unlimited	 Target RedCard (Mastercard)
Credit Line	\$8,000	\$8,000
Statement Balance	\$0	\$0
Purchase APR	24.86%	29.95%
Rewards	• 3% cash back on dining at restaurants, including takeout and eligible delivery services. • 3% cash back on drugstore purchases. • 5% cash back on travel purchased through Chase Travel. • 1.5% cash back on all other purchases.	• 5% off every day at Target and Target.com. Earn rewards toward a Target gift card: • 2% in rewards on dining and gas purchases. • 1% in rewards everywhere else outside of Target.

Results and discussion

As preregistered, we compared the proportion of participants choosing the reward-maximizing card between the Co-branded group and the Control group. As predicted, significantly fewer participants in the Co-branded group chose to use the reward-maximizing

credit card (co-branded with Walmart or Target) than in the Control group (25% vs. 52%; $\chi^2(1) = 28.7, p < .001$). In other words, consistent with the previous studies, people were less likely to use the reward-maximizing credit card if it was co-branded.

After running the study, we noticed that there was ambiguity in our stimuli about whether the fuel reward of the Walmart Rewards Mastercard applied to all fuel stations or only to Walmart fuel stations. There was thus a possibility that the effect was driven by a portion of Walmart Rewards Mastercard users who correctly understood that the 2% fuel reward only applied to Walmart fuel stations. To address this, we looked into each co-branded credit card separately to understand whether the effect was mainly driven by the Walmart Rewards Mastercard users, although we did not preregister this analysis. It turned out that there was a strongly significant effect when testing only the Target RedCard Mastercard and a marginally significant effect when testing only the Walmart Rewards Mastercard: 14% of those with Target RedCard Mastercard (18 out of 121) chose to use it (vs. 52% selecting Discover; $\chi^2(1) = 41.4, p < .001$), and 39% of those with the Walmart Rewards Mastercard (36 out of 92) chose to use it (vs. 52% selecting Discover; $\chi^2(1) = 3.7, p = .055$).

As noted earlier, the Target rewards could only be used at Target like gift card dollars. One might wonder whether the large effect with the Target RedCard Mastercard was driven by users devaluing Target gift card dollars. However, we postulated that people who chose to have the Target RedCard Mastercard were likely frequent Target shoppers and thus would value the Target-only rewards about the same as cash. Indeed, 79% of the participants with the Target RedCard Mastercard indicated that, to them, a dollar in Target store credit was worth no less than a dollar in cash.

Overall, these results replicated our previous findings with the co-branded credit cards that participants actually possessed in their real lives, arguing against lack of brand preference as an explanation for the co-branding effects we see in this paper.

10. Mouse-cursor movement data for the Visa card in Study 3

Table S9. Number of Visits and Total Visiting Time for Each Visa Cell

	Mean (SD)		<i>t</i> (587)	<i>p</i>
	Control condition	Co-branded condition		
Frequency of opening the cell				
Credit Line	1.64 (0.83)	1.60 (0.91)	0.62	.491
Current Balance	1.64 (0.88)	1.53 (0.88)	1.44	.150
APR	1.63 (0.87)	1.70 (1.04)	-0.95	.344
Rewards	4.23 (3.40)	3.47 (1.84)	3.36	< .001
Total time spent at the cell (seconds)				
Credit Line	1.18 (0.94)	1.15 (0.87)	.29	.775
Current Balance	0.87 (0.88)	0.84 (0.88)	.81	.421
APR	1.00 (0.65)	1.00 (0.59)	-0.11	.914
Rewards	6.20 (3.79)	5.32 (2.93)	3.14	.002

Table S9 summarizes the mouse cursor movements over the Visa card for both conditions. The Visa card itself remained unchanged between conditions. However, it was juxtaposed with the Mastercard that was manipulated to be either co-branded with REI or not co-branded. We have proposed that when a credit card features a merchant brand, users may assume that the co-branded credit card mainly rewards purchases within the featured brand. Thus, when the Mastercard was co-branded with REI, they were less likely to carefully compare the Visa card's rewards to the Mastercard. Instead, they might simply assume that the REI Mastercard mainly rewarded REI purchases, while the Visa card was better for general spending.

Consistent with this proposition, when the Mastercard was co-branded with REI, participants spent less time looking at the Visa card's rewards because the presence of the REI brand reduced their willingness to carefully compare the Visa card's rewards to those of the REI Mastercard.

11. Study 3: Gazing time after excluding outlier hovers beyond 3 SDs

Table S10. Gazing Time for each Mastercard cell in Study 3 After Excluding Outliers beyond 3 Standard Deviations

	Mean (SD)		$t(587)^2$	p
	Control condition	Co-branded condition		
Credit Line	0.81 (0.49)	0.82 (0.49)	-.18	.858
Current Balance	0.71 (0.44)	0.72 (0.39)	-.23	.820
APR	0.81 (0.42)	0.81 (0.46)	-0.14	.888
Rewards	5.61 (3.93)	4.41 (2.89)	4.20	<.001

12. Unobserved factors precluding us from quantifying the impact on banks

For credit card-issuing banks, revenues from credit card products primarily come from two sources: interchange fees charged on transactions and interest charged on revolving balances. However, issuing banks such as Chase and Citi typically do not publicly disclose their interchange rates. In addition, we do not observe whether participants carried revolving balances, nor the size of those balances. The revenue structure is further complicated for co-branded credit cards, where issuing banks often share a portion of credit card revenues with merchant partners, and the terms of such agreements are generally confidential.

It is even more difficult to estimate the variable costs that issuing banks incur at the transaction level. These costs may include, but are not limited to, rewards expenses, borrowing costs, credit losses, fraud and dispute costs, network fees, and processing fees. Because the profitability of a credit card product depends on both revenues and costs, and because both are difficult to measure with unobserved information, it is challenging to quantify the impact of co-branded credit card underuse on issuing banks. Nevertheless, in the manuscript, we identify the likely sources of several key impacts and discuss actions firms can take to mitigate the underuse of their co-branded credit card products.

² The d.f. remains the same between the 2-SD and 3-SD exclusions because, as preregistered, we excluded outlier mouse cursor hovers (i.e., gazes), pooling across all individuals, instead of excluding individuals. After the exclusion, individuals without a valid gaze were coded as 0, indicating that they did not take a meaningful look at the information.